

## **Tag attachment for full-annual cycle tracking of Snow Bunting**

### **1. Background**

This protocol supplements the general passerine tag attachment protocols and describes considerations specific to Snow Bunting (*Plectrophenax nivalis*). For general guidance on leg-loop harness construction and application, refer to the [Figure-8 Leg Loop Harness protocol](#). This document focuses on modifications and recommendations for long-term (>1 year) tracking of Snow Bunting.

Leg-loop harnesses have been used successfully to retain tracking devices for more than one year, although retention depends on harness construction and materials. Short-term deployments (up to approximately 2 month) are typically achieved using elastic sewing thread, whereas longer-term deployments use elastic jewelry cord (e.g., Stretch Magic). Tags with eyelets or tubes are preferred for studies that require tags to stay on indefinitely, or used in studies where each harness may need to be fitted individually. Accordingly, this protocol focuses on adjustable crimped harnesses used with eyelet- or tube-mounted transmitters.

Snow Bunting undergo some of the largest seasonal changes in body mass documented in passerines. Consequently, harness fit should account not only for the bird's current body condition but also for expected changes in body mass over the deployment period. For instance, Snow Bunting carrying high fat loads (fat score 4–6) require a tighter initial harness fit so that the transmitter remains properly positioned after fat reserves are depleted. Conversely, lean birds (fat score 0–2) require slightly more slack to accommodate subsequent fat deposition. Birds with extremely high fat scores (6–7) may be difficult to fit because fat deposits around the abdomen and thighs can make it challenging to position the leg loops correctly behind the knees.

### **2. Considerations**

Adjustable harnesses with eyelet or tubed tags can be fit to a bird in the field. Applying adjustable harness in the field is most easily performed with two people and can take up to five minutes.

Migratory Bird Permits (CWS) are required, with specific permissions related to attaching tracking devices to birds. We strongly recommend that the fitter practice deploying this type of harness before attempting it on a live bird.

### **3. Purpose**

To provide instructions for the preparation and deployment of adjustable leg-loop harnesses for long-term attachment of tracking devices to Snow Bunting.

### **4. Materials**

- 0.7 mm black jewelry cord (e.g. Stretch Magic)
- Silver crimp beads
- Crimping tool
- Crochet hook
- Gel-style super glue (e.g. LePage Ultragel Super Glue)
- Scissors (with blunt end)
- Ruler
- Baking soda
- Eyelet or tubed tags
- Tag activator
- SRX Receiver

## **5. Methods**

### **5a. Harness preparation**

- Activate tag using tag activator
- Ensure tag is activated by confirming manufacturer ID on SRX Receiver
- Determine the approximate leg loop size for your species.
- Cut thread at an angle (so that it will easily slip into the crimps), leaving a length that leaves an extra 10-20 cm on either end.
- Insert the elastic thread in the anterior tube or eyelets of the transmitter.
- When the tube is positioned halfway across the thread, return the thread through the anterior tube and tighten the loop.
- Apply glue to the thread on the anterior tube. Quick dry with baking soda. This prevents the tag from moving across the thread.
- Pass one end of the thread through the posterior tube of the transmitter.
- Insert the second end of the thread through the opposite side of the posterior tube of the transmitter.
- Pass each end of the thread through two crimps.
- Half tighten the crimps to prevent the crimps from sliding off when deploying the transmitter on the bird while in the field.

### **5b. Harness application**

- Hold the bird in the bander's grip
- Place the loop over the bird's foot and over/around the thigh. You can use a crochet hook to assist you in manipulating the thread.
- While holding the tension, position the tag on the bird's back.
- While holding the tag on the bird's back, place the second loop over the bird's other leg. You can use a crochet hook to assist you in manipulating the thread.

- Have a second person help you pull both ends of stretch magic through the crimp bead to tighten the harness.
- Once the desired harness tension is reached, maintain tension on the cord while checking that the loops are correctly positioned around the hips. Ensure the loops rest around the thighs between the legs and body and cannot slip down the legs or restrict normal movement.
- Remove any feathers trapped beneath the harness before making the final adjustment.
- Check the tension by lifting the bottom of the tag (where the antenna joins the tag body). If you cannot move it at all, it might be too tight and restrict movement or blood flow. If the harness pulls one or both of the bird's legs out from under the body, it is too tight. If you can flip the tag, it is too loose. If you can easily slide the tag forward and backward, or side to side, without stretching the harness, it is probably too loose.
- Confirm that both legs move freely and that the bird can perch normally on your finger.
- Once the fit is satisfactory, have the second helper position the crimp bead immediately adjacent to the tag and crimp it once. Rotate the crimping tool approximately one-quarter turn and crimp again. The finished crimp should appear flattened.
- Apply a second crimp directly behind the first one to add increased strength. Gently pull on the cord to verify that the attachment is secure.
- Apply a small amount of gel superglue to the crimp bead to increase strength, then apply baking soda to accelerate curing.
- Trim the excess cord with scissors.

### **5c. Ensure proper fit, and release bird**

- Check that the feet are able to perch on your fingers, and that the wings are free.
- When holding the bird in the photographer's grip, the tag should sit high on the bird's back in line with their spine, free of feathers (if solar-powered, or has a light sensor), with the antenna falling straight down the bird's tail. The bird will tend to preen them into place and cover them with feathers over time.
- Tug lightly on the tag to check movement: if you can easily slide the tag forward and backward, or side to side, without stretching the harness, it is probably too loose. If the harness pulls one or both of the bird's legs out from under the body, it is too tight.
- Release the bird very low to the ground, in case it refuses to fly. On occasion, passerines may initially refuse to fly, but will resume normal behaviour after some time passes. The newfound pressure on their back will sometimes cause the birds to resist flying. If the tag is too tight, they'll feel more pressure on their back.
- Watch the bird to ensure it acclimates to its tag after release; the bird can be left once it appears to have returned to normal behaviour.
- If the bird refuses to fly within a minute or two, the tag should be removed by carefully cutting the leg loops. One person should hold the bird, while another pulls gently on the tag to create a gap where the scissors can be safely maneuvered to cut the harness near the tag. The bird can then be released as described above.